Unit Test Report - Optional Comments (optionalCommentsOn)

Date: 2026-11-11 | Repo: reactive_forms, branch feature/optional-comments-on | Runner: flutter test

Result: 48 of 48 cases passed, 0 failed.
Rules under test: `requireCommentsOn` = comment shown and mandatory; `optionalCommentsOn` = comment shown, not mandatory; if an answer is in both, mandatory wins; `disableCommentsOn` hides the comment and has the highest priority; with no `optionalCommentsOn` the behaviour is unchanged.
Every row was copied from the actual test run output. "Input" is the question config / answer / comment; "Expected" is what the test asserted; "Received" is what the code returned.

A. Renderer rules (reactive_forms) - test/optional_comments_test.dart (42 cases)

ID	Case	Question config	Input	Expected	Received	Result
E1a	show comment on No	`legacy`	requireCommentsOn=[No], answer=No	`true`	`true`	**PASS**
E1b	hide comment on Yes	`legacy`	requireCommentsOn=[No], answer=Yes	`false`	`false`	**PASS**
E2a	empty comment invalid	`legacy`	answer=No, comment=""	`false`	`false`	**PASS**
E2b	filled comment valid	`legacy`	answer=No, comment="ok"	`true`	`true`	**PASS**
E2c	not optional	`legacy`	answer=No	`false`	`false`	**PASS**
E2d	no optional config	`legacy`	legacy field	`false`	`false`	**PASS**
E3a	show on Yes	`noLists`	no lists, answer=Yes	`true`	`true`	**PASS**
E3b	show on No	`noLists`	no lists, answer=No	`true`	`true`	**PASS**
E3c	comment control is mandatory	`noLists`	no lists, empty comment	`false`	`false`	**PASS**
E4	no _comment control	`noComments`	hasComments=false	`false`	`false`	**PASS**
E5a	hidden on disabled answer	`(inline) disableCommentsOn=[No]`	disableCommentsOn=[No], No	`false`	`false`	**PASS**
E5b	shown on other answer	`(inline) disableCommentsOn=[No]`	disableCommentsOn=[No], Yes	`true`	`true`	**PASS**
N1a	shown on Yes	`optBoth`	optionalCommentsOn=[Yes,No], Yes	`true`	`true`	**PASS**
N1b	shown on No	`optBoth`	optionalCommentsOn=[Yes,No], No	`true`	`true`	**PASS**
N1c	empty comment is valid on Yes	`optBoth`	answer=Yes, comment=""	`true`	`true`	**PASS**
N1d	empty comment is valid on No	`optBoth`	answer=No, comment=""	`true`	`true`	**PASS**
N1e	isCommentOptional on Yes	`optBoth`	Yes	`true`	`true`	**PASS**
N1f	validateFieldCommentsIfRequired passes when empty	`optBoth`	answer=No, comment=""	`true`	`true`	**PASS**
N2a	Yes: shown	`mixed`	require=[No], optional=[Yes], Yes	`true`	`true`	**PASS**
N2b	Yes: empty comment valid	`mixed`	answer=Yes, comment=""	`true`	`true`	**PASS**
N2c	No: shown	`mixed`	require=[No], optional=[Yes], No	`true`	`true`	**PASS**
N2d	No: empty comment invalid	`mixed`	answer=No, comment=""	`false`	`false`	**PASS**
N2e	No: validateFieldCommentsIfRequired fails	`mixed`	answer=No, comment=""	`false`	`false`	**PASS**
N2f	No: filled comment valid	`mixed`	answer=No, comment="reason"	`true`	`true`	**PASS**

N3	valid state follows answer (No->Yes->No), empty comment	`mixed`	No, Yes, No	`false,true,false`	`false,true,false`	**PASS**
N4a	not optional	`both`	require=[No], optional=[Yes,No], No	`false`	`false`	**PASS**
N4b	empty comment invalid	`both`	answer=No, comment=""	`false`	`false`	**PASS**
N4c	Yes still optional	`both`	answer=Yes	`true`	`true`	**PASS**
N5a	hidden on No	`disabled`	optional=[Yes], disable=[No], No	`false`	`false`	**PASS**
N5b	shown on Yes	`disabled`	optional=[Yes], disable=[No], Yes	`true`	`true`	**PASS**
N5c	Yes: empty comment valid	`disabled`	answer=Yes, comment=""	`true`	`true`	**PASS**
N6a	shown on Yes	`optOnlyYes`	optional=[Yes], require=[], Yes	`true`	`true`	**PASS**
N6b	hidden on No (no fallback)	`optOnlyYes`	optional=[Yes], require=[], No	`false`	`false`	**PASS**
N7a	shown on Yes	`sheetStyle`	require=[''], optional=[Yes,No], Yes	`true`	`true`	**PASS**
N7b	blank entry never matches	`sheetStyle`	require=[''], answer=''	`false`	`false`	**PASS**
N7c	empty comment valid on No	`sheetStyle`	answer=No, comment=""	`true`	`true`	**PASS**
N7d	comma string " Yes , " is trimmed	`optYesStr`	optional=" Yes , ", Yes	`true`	`true`	**PASS**
N7e	comma string: No not optional	`optYesStr`	optional=" Yes , ", No	`false`	`false`	**PASS**
N8a	optional if a selected answer is optional	`optOnlyYes`	optional=[Yes], value=[Yes,No]	`true`	`true`	**PASS**
N8b	not optional if none selected is optional	`optOnlyYes`	optional=[Yes], value=[No]	`false`	`false`	**PASS**
N9a	null answer is not optional	`optBoth`	value=null	`false`	`false`	**PASS**
N9b	empty answer is not optional	`optBoth`	value=''	`false`	`false`	**PASS**

B. Booking app parsing - test/checklist_optional_comments_test.dart (6 cases)

ID	Case	Input	Expected	Received	Result
B1	comma list parsed	optionalCommentsOn='Yes,No'	`[Yes, No]`	`[Yes, No]`	**PASS**
B2	spaces trimmed	optionalCommentsOn=' Yes , No '	`[Yes, No]`	`[Yes, No]`	**PASS**
B3	blank string gives empty list	optionalCommentsOn=''	`[]`	`[]`	**PASS**
B4	missing key gives empty list	key absent	`[]`	`[]`	**PASS**
B5	requireCommentsOn unchanged	requireCommentsOn='No'	`[No]`	`[No]`	**PASS**
B6	toJson carries optionalCommentsOn	optionalCommentsOn='Yes'	`[Yes]`	`[Yes]`	**PASS**

C. Question configurations used in section A

legacy

```
{
 "type": "radio",
 "options": [
  "Yes",
  "No"
 ],
 "required": true,
 "hasComments": true,
 "requireCommentsOn": [
  "No"
 ]
}
```

optBoth

```
{
 "type": "radio",
 "options": [
  "Yes",
  "No"
 ],
 "required": true,
 "hasComments": true,
 "requireCommentsOn": [],
 "optionalCommentsOn": [
  "Yes",
  "No"
 ]
}
```

mixed

```
{
 "type": "radio",
 "options": [
  "Yes",
  "No"
 ],
 "required": true,
 "hasComments": true,
 "requireCommentsOn": [
  "No"
 ],
 "optionalCommentsOn": [
  "Yes"
 ]
}
```

both

```
{
 "type": "radio",
 "options": [
  "Yes",
  "No"
 ],
 "required": true,
 "hasComments": true,
 "requireCommentsOn": [
  "No"
 ],
 "optionalCommentsOn": [
  "Yes",
  "No"
 ]
}
```

disabled

```
{
 "type": "radio",
 "options": [
  "Yes",
  "No"
 ],
 "required": true,
 "hasComments": true,
 "optionalCommentsOn": [
  "Yes"
 ],
 "disableCommentsOn": [
  "No"
 ]
}
```

optOnlyYes

```
{
 "type": "radio",
 "options": [
  "Yes",
  "No"
 ],
 "required": true,
 "hasComments": true,
 "requireCommentsOn": [],
 "optionalCommentsOn": [
  "Yes"
 ]
}
```

sheetStyle

```
{
 "type": "radio",
 "options": [
  "Yes",
  "No"
 ],
 "required": true,
 "hasComments": true,
 "requireCommentsOn": [
  ""
 ],
 "disableCommentsOn": [
  ""
 ],
 "optionalCommentsOn": [
  "Yes",
  "No"
 ]
}
```

noLists

```
{
 "type": "radio",
 "options": [
  "Yes",
  "No"
 ],
 "required": true,
 "hasComments": true
}
```

noComments

```
{
 "type": "radio",
 "options": [
  "Yes",
  "No"
 ],
 "required": true,
 "hasComments": false
}
```

optYesStr

```
{
 "type": "radio",
 "options": [
  "Yes",
  "No"
 ],
 "required": true,
 "hasComments": true,
 "optionalCommentsOn": " Yes , "
}
```

D. Not covered (manual device testing needed)

- Red asterisk showing/hiding next to the comment label as the answer changes.
- "Next" / "Submit" blocking or allowing in the real UI; only validateFieldCommentsIfRequired is covered.
- Hidden-comment text being dropped from submitted data.

- Grouped questions, duplicated question sets, and multiselect/number/temp/file types with comments.
- End-to-end flow of the optionalCommentsOn column from the Google Sheet through the backend to the app.
- Pre-existing repo tests were not run.